

Credit Card Customer Behavior

Hypothesis Testing + Clustering Extension

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September 01, 2026

Project Overview

Research Question

Do distinct patterns of credit card usage carry statistically measurable differences, and can we identify natural customer profiles through clustering?

	Part 1	Part 2
Approach	Hypothesis Testing	K-Means Clustering
Tools	<code>t.test()</code> , <code>cor.test()</code>	<code>kmeans()</code> , <code>dplyr</code>
Goal	Confirm or debunk claims	Discover customer groups
Dataset	CC GENERAL — 8,950 customers, 18 variables	

Significance level for all tests: $\alpha = 0.05$

Data Wrangling with dplyr

```
cc_clean <- cc %>%  
  filter(!is.na(CREDIT_LIMIT), !is.na(MINIMUM_PAYMENTS)) %>%  
  mutate(  
    CA_Group = if_else(CASH_ADVANCE_FREQUENCY >= 0.25,  
                       "Heavy CA", "Light CA"),  
    LOG_BALANCE = log1p(BALANCE)  
  )
```

Rows after cleaning: 8636

Heavy CA	Light CA
2204	6432

Part 1: Hypothesis Testing — Overview

H1 – Heavy Cash-Advance Users Carry Higher Balances

Test: One-tailed Welch's two-sample t -test

Expected: Confirmed

H2 – Balance Strongly Predicts Credit Limit (Debunk)

Test: Pearson correlation `cor.test()`

Expected: Debunked

	H1	H2
Null Hypothesis	$\mu_{\text{Heavy}} = \mu_{\text{Light}}$	$\rho = 0$
Alternative	$\mu_{\text{Heavy}} > \mu_{\text{Light}}$	$\rho \neq 0$ (strong)
Key metric	t -statistic, p -value	r, r^2

H1: Formal Statement

H_0 : $\mu_{\text{Heavy CA}} = \mu_{\text{Light CA}}$ ($\mu_{\text{Heavy CA}}$ is the average balance of Heavy CA users, and it is equal to $\mu_{\text{Light CA}}$ which is the average balance of Light CA users)

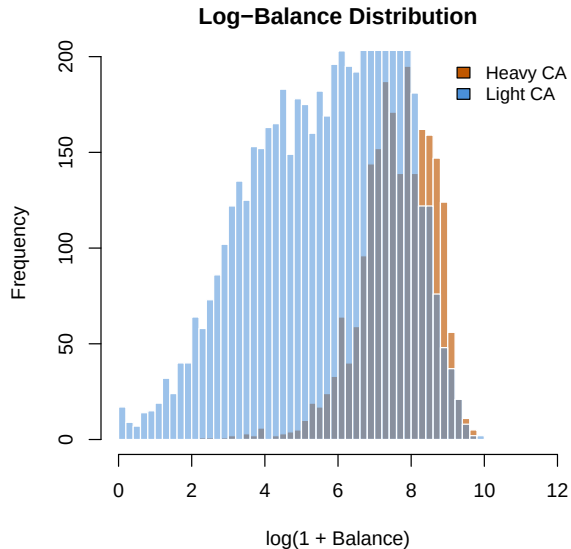
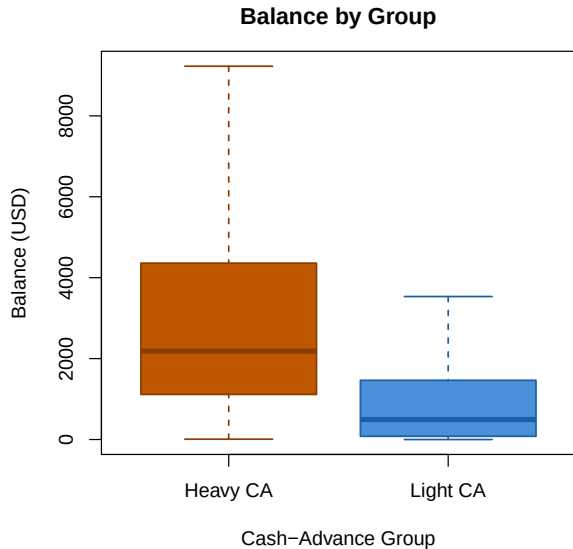
H_1 : $\mu_{\text{Heavy CA}} > \mu_{\text{Light CA}}$ ($\mu_{\text{Heavy CA}}$ is the average balance of Heavy CA users, and it is greater than $\mu_{\text{Light CA}}$ which is the average balance of Light CA users)

Rationale

The correlation matrix shows $r(\text{BALANCE}, \text{CASH_ADVANCE_FREQUENCY}) \approx 0.449$. Customers who rely on cash advances tend to accumulate higher account balances (BALANCE = balance amount left in account to make purchases) — because cash advances incur high fees and interest immediately upon withdrawal.

Test chosen: Welch's two-sample t -test (`var.equal = FALSE`) — safe default when group variances differ.

H1: Exploratory Plots



H1: Welch's t -Test

```
t_result_h1 <- t.test(  
  x      = heavy,  
  y      = light,  
  alternative = "greater",  # one-tailed: Heavy > Light  
  var.equal  = FALSE      # Welch's t-test  
)
```

t = 32.319

df = 2889.6

p = 3.43e-196

95% CI (lower bound): \$1775

H1: Decision & Interpretation

t-statistic : 32.319

p-value : 3.43e-196

95% CI lower bound: \$1775.11

Decision : REJECT H0

Interpretation

The large positive t and the near-zero p -value provide strong statistical evidence: heavy cash-advance users carry **significantly higher** balances on average. CA frequency is a **measurable financial risk signal**.

H2 (Debunk): The Intuitive Claim

Common Intuition

“Customers with higher balances must have higher credit limits — the bank gave them more credit, so they use more of it.”

$H_0 : \rho = 0$ (ρ is the correlation between BALANCE and CREDIT_LIMIT, and it is equal to zero, meaning there is no relationship)

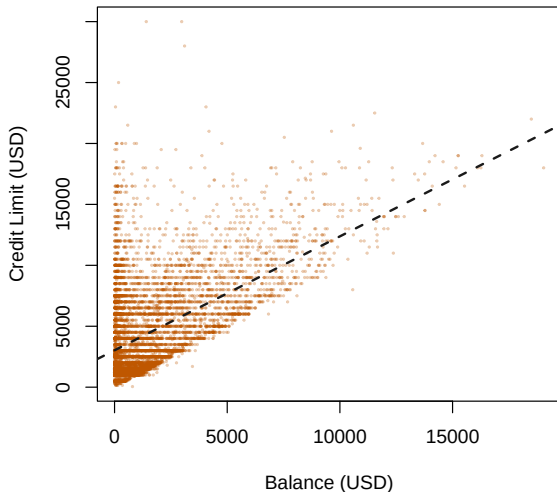
$H_1 : \rho \neq 0$ (ρ is the correlation between BALANCE and CREDIT_LIMIT, and it is not equal to zero, meaning there is a strong positive relationship)

Debunk strategy — three prongs:

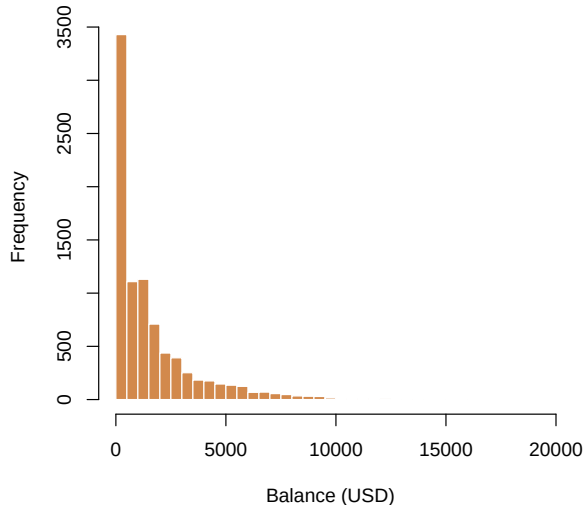
- 1 **Moderate** r from `cor.test()` — r^2 explains only $\approx 29\%$ of variance
- 2 **Severe outliers** in `boxplot()` that distort the visual pattern
- 3 **Right-skewed distribution** in `hist()` — normality assumption violated

H2: Scatterplot & Distribution

Balance vs. Credit Limit

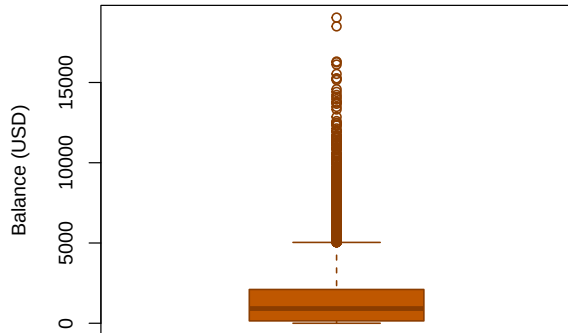


Balance – Highly Right-Skewed

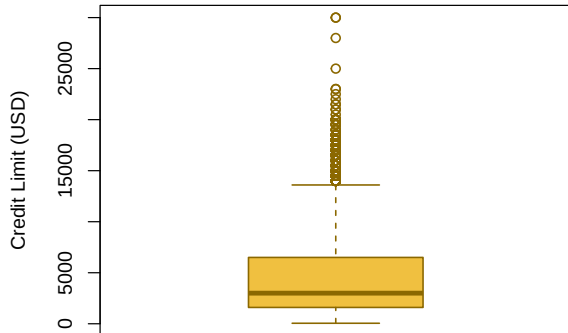


H2: Outlier Detection

Balance – Outliers



Credit Limit – Distribution



Balance outliers (IQR rule): 666 customers

H2: Pearson Correlation Test & Debunk

```
cor_h2 <- cor.test(cc_clean$BALANCE, cc_clean$CREDIT_LIMIT,  
                   method = "pearson")
```

r (Pearson) : 0.5355

r^2 (variance expl.): 28.68%

p-value : 0.00e+00

$|r| < 0.35?$: NO -> DEBUNKED

Key Lesson: Significance $\neq$ Practical Strength

With $n = 8,950$, even a moderate correlation gives a near-zero p -value. The actual $r \approx 0.54$ seems moderate, but $r^2 \approx 0.29$ means BALANCE explains only $\approx 29\%$ of CREDIT_LIMIT variance. Banks set limits *prospectively* (based on income and credit history), not in response to current balance — the strong-prediction claim is **false**.

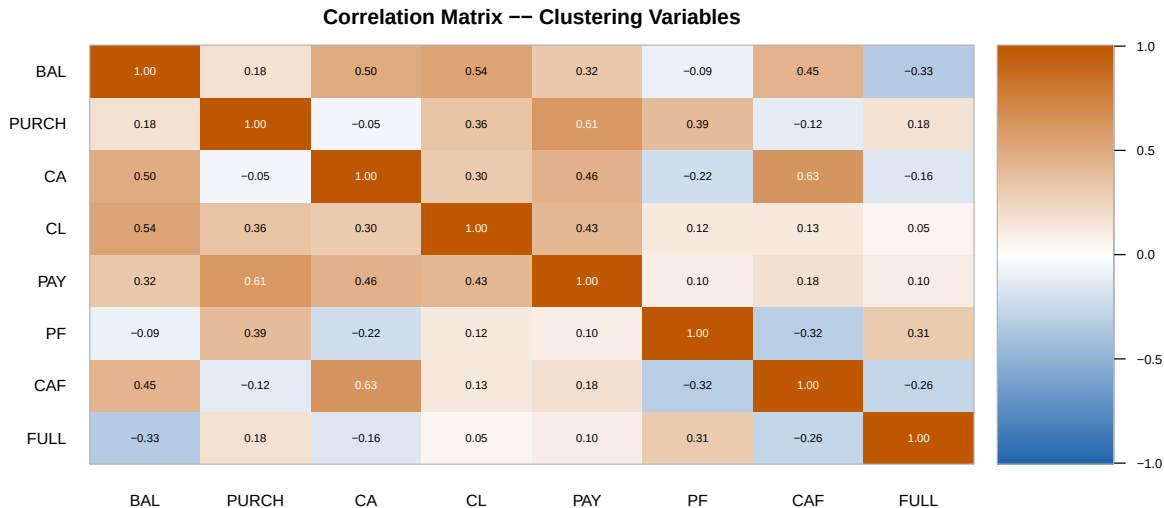
Key Correlations Found

Three most important relationships

- **PURCHASES and PAYMENTS** $r \approx 0.60$ (strong positive) — customers who spend more also tend to pay more.
- **CASH_ADVANCE and BALANCE** $r \approx 0.50$ (moderate positive) — heavier cash-advance use is associated with a higher account balance.
- **PRCFULLPAYMENT and BALANCE** $r \approx -0.32$ (negative) — customers who pay in full carry much lower account balances.

The matrix also revealed high redundancy between some pairs (e.g., PURCHASES and ONEOFF_PURCHASES: $r \approx 0.92$), which guided the selection of non-overlapping variables for clustering.

Correlation Matrix



Complementary Question:

The hypothesis tests confirmed specific claims about individual variables.

Now we ask: *Can we find natural customer groups in the data?*

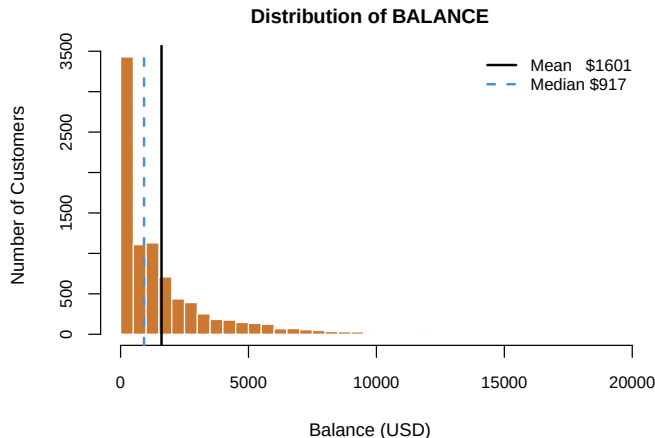
Method: K-Means Clustering — an exploratory, unsupervised technique

Input: 8 financial variables, standardised

Result: 5 interpretable customer profiles

EDA: Key Findings

- **8,950 customers**, 18 variables
- All major financial variables are **strongly right-skewed**:
 - PURCHASES: skew = 8.14
 - CASH_ADVANCE: skew = 5.17
 - BALANCE: skew = 2.39
- Mean $\gg$ Median for most variables — a small group of high-activity customers pulls the average up
- Extreme values **kept** — they represent real, meaningful customer behavior



Why K-Means Clustering?

The dataset has no predefined customer categories

The EDA showed that customers behave very differently from one another. Clustering groups observations that share similar characteristics, without requiring a predefined label.

- **8 variables** selected: BALANCE, PURCHASES, CASH_ADVANCE, CREDIT_LIMIT, PAYMENTS, PURCHASES_FREQUENCY, CASH_ADVANCE_FREQUENCY, PRCFULLPAYMENT
- Variables were **standardised** so that dollar-valued variables do not dominate frequency variables (0 to 1 scale)
- **K = 5** chosen by comparing several values — five groups provide a useful balance between separation and interpretability
- K-Means is used here as an **exploratory complement** to the hypothesis-testing analysis above

K-Means Result: 5 Customer Groups

Cluster Name	% Customers	Key Characteristics
Low-Activity Customers	38.7%	Very low purchases, low purchase frequency
Regular Purchasers	32.7%	Frequent purchases, very low cash-advance use
Full-Payment Customers	14.8%	Pays full balance consistently, lowest debt carried
High Cash-Advance Users	12.4%	Very high balance driven by cash advances
High-Value Customers	1.3%	Extremely high spending and credit limit (n = 112)

Conclusions

Part 1: Hypothesis Testing

- **H1 Confirmed** — Heavy cash-advance users carry significantly higher balances ($p \ll 0.05$). CASH_ADVANCE_FREQUENCY is a reliable credit-risk signal.
- **H2 Debunked** — Despite $p < 0.05$, BALANCE explains only $\approx 29\%$ of CREDIT_LIMIT variance ($r \approx 0.54$, $r^2 \approx 0.29$). Statistical significance does not imply practical strength.

Part 2: Clustering Extension

K-Means identified **five interpretable customer profiles** that summarise different financial behaviors across the 8,950 customers. These are **exploratory profiles, not official categories** — they complement and contextualise the hypothesis-testing findings above.

References

- **Dataset:** CC GENERAL — Kaggle Credit Card Customer Segmentation (8,950 obs.)
- **R Packages:** dplyr (wrangling) · Base R (stats, graphics)
- **Statistical Functions:** `t.test()` · `cor.test()` · `kmeans()`
- **Visualization:** `boxplot()` · `hist()` · `plot()` · `barplot()` · `image()`
- **Concepts Applied:**
 - Welch's Two-Sample t -Test
 - Pearson Correlation — Effect Size vs. Significance
 - K-Means Clustering ($K = 5$, standardised variables)
 - Outlier Detection via IQR Rule